Bench exercise: a dovetail that drops in

Japanese Joinery by Hand · Episode 5. Before cutting wood, build the *koshikake ari-tsugi* 腰掛蟻継ぎ in stiff card. You'll see for yourself why the male goes in from above, and why it can't slide apart. No sharp tools beyond a craft knife and a straightedge. About 45 minutes.

1 · Draw it full size

1. On card, draw a "sill" 105 mm wide (the plan view), long enough to hold the joint: about 300 mm.
2. Draw a center line down its length.
3. Across it, draw the shoulder line. From the shoulder, draw a dovetail 50 mm long, with its neck 30 mm and its head 50 mm, laid out evenly either side of the center line. That's the episode's drawing: each side slopes about 1 in 5.
4. Check it against the ranges from the engineering study: the neck is ¼–⅓ of the width (26–35 mm), and the length ¼–½ of the width (26–52 mm).

2 · Cut two layers

1. Cut the male's dovetail tongue and the female's matching socket from two pieces of card. These are the upper halves.
2. Cut two plain strips for the lower halves. On the female's strip, leave a short ledge sticking out past the socket line: that's the seat (*koshikake* 腰掛).
3. Glue each upper half onto its lower half.

3 · Try every way in

1. Try sliding the male in from the end, along the sill. What stops it?
2. Try sliding it in from the side. What stops it?
3. Now lower it straight down from above. The dovetail settles into its socket, and the male's cut-back underside lands on the seat.
4. Pull the two pieces apart along their length. The flare hooks them together. Now lift the male: it comes straight out.

4 · Think about the house

- Which piece would the anchor bolt go in? (The upper piece, beside the joint.)
- Where along the sill would you put this splice? (Away from posts and underfloor vents.)

What you should have

- A card joint that only goes together one way, and a clear sense of why.

Proportions vary; the 50/50/30 drawing is the episode's, inside one engineering study's ranges, not a standard. Next episode: marking and cutting it in wood.